

A Comprehensive Survey on Robust Image Watermarking

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ABSTRACT

With the rapid development and popularity of the Internet, multimedia security has become a general essential concern. Especially, as manipulation of digital images gets much easier, the challenges it brings to authentication certification are increasing. As part of the solution, digital watermarking has made significant contributions to image content security and has attracted increasing attention. In this paper, we present a comprehensive review on digital image watermarking methods that were published in recent years illustrating the conventional schemes in different domains. We provide an overview of geometric invariant techniques and emerging watermarking methods for novel medias, such as depth image based rendering (DIBR), high dynamic range (HDR), screen content images (SCIs), and point cloud model. Particularly, as deep learning has achieved a great success in the field of image processing, and has also successfully been used in the field of digital watermarking, learning-based watermarking methods using various neural networks are summarized according to the utilization of neural networks in the single stage training (SST) and double stage training (DST). Finally, we provide an analysis and summary on those methods, and suggest some future research directions.

1. Introduction

The development of technologies makes it possible to represent the world around us and save it in a certain digital forms, such as image [1], audio [2], video [3, 4], and text [5]. Digital media has become an inseparable part of our daily life, which makes it convenient to record activities, express and share emotion with each other [6]. The easy-to-use technologies [7], allow the non-professional users to manipulate the multimedia data easily [8], such as the editing tool PhotoShop [9].

On the one side, with the perspective of information hiding [10], much more information can be embedded within the image purposely. The digital media nowadays that usually has higher a resolution needs a huge memory space to store. The fact, however, is that there are amounts of redundancy in the digital media, which can be removed for time-saving. Meanwhile, it offers an opportunity to hide information by modifying the image information directly [11]. In particular, in the military, commercial, financial, secret agreements and contracts between the negotiating parties, it is of great importance to dispatch amount of messages [12] through the Internet in the form of digital media. On the other side, it also has a potential risk of fake media [13], which is hard to be identified by bare human eyes or simple detection methods. This thus increases the threat of intellectual property infringement [14–16] to verify the ownership or realness of the media. Therefore, how to effectively prove the copyright ownership of data transmitted on the Internet becomes particularly important in the presence of conflict of

benefits [17].

On the purpose of protecting the content security of digital media, many techniques have been proposed for different applications [18], such as copyright protection, fingerprinting, and authentication. In terms of copyright protection and authentication, cryptography, steganography and watermarking have been accepted as the effective and practical techniques to protect the copyright of digital multimedia [19]. Cryptography technology is the most effective way to protect products. However, only authorized decryption targets can store and use data of the encrypted products [20]. Once the password of the data is cracked by an unauthorized person in some way, the product will be controlled by the cracker, and the original user has no way to track the copying and secondary dissemination of the product. Different from the cryptography that the encrypt operation is know for everyone, the main idea of the steganography is that the attacker is unaware of the secret communication between the sender and receiver, and the stego itself is completely undetectable [21]. However, the information can be eliminated or disturbed easily if the attacker manipulates the media maliciously. Subsequently, the emergence of digital watermarking technology perfectly solves the shortcomings of cryptography and other technical protection [22–24]. It is a technique similar to steganography but overcomes the weakness existed in both cryptography and steganography. If the carriers were tampered, the owners can prove the ownership of digital medias by extracting the embedded watermark.

In the past decades, various watermarking techniques have been proposed. Some comprehensive overview papers of digital watermarking have been presented in many aspects. Saini and Shrivastava [25] and Shukla et al. [26] give a brief introduction of the properties, classification, and per-

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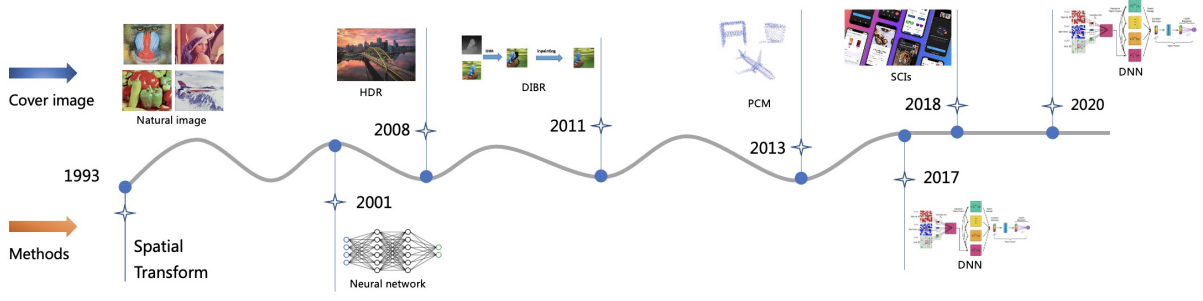


Figure 1: The story line of the survey.

formance analysis of digital watermarking. Dixit et al. [27] introduce the techniques of watermarking with different embedding domain. Begum and Uddin [28] detail the watermarking according to different applications and embedding domains. Kumar et al. [29] and Singh et al. [30] illustrate the techniques in e-governance and designed using soft computing techniques respectively. Yet, these works focus on the watermarking techniques deriving from handcraft features, while the deep-feature-based digital watermarking methods have been also used and have shown satisfied results.

In this paper, we intend to make a comprehensive review of the recent researches on *robust* watermarking technologies, leaving aside other applications, e.g., semi-fragile watermarking [31] and fragile watermarking [32]. To start with, we enumerate the properties that are required for digital watermarking methods. Then, following the time-line in Fig. 1, we present the watermarking works in two directions: the change of cover image and the evolution of the means of watermarking. To be more specific, the main contributions are summarized as follows:

- Initially, we start the paper by presenting a thorough review of robust watermarking works for classic natural images. Typically, we introduce watermarking methods by following a general watermark processing, e.g., spatial domain and transform domain.
- Subsequently, we present watermarking methods for two kinds of complicated images and a rendered image proposed so far, in particular, HDR, SCIs, and DIBR. We illustrate the main concerns for each kind of images and the corresponding solutions to these issues.
- Then, we systematically review the learning-based image watermarking methods. In particular, we first give a priori impression about information hiding by introducing some related steganography works using deep neural networks. Then we present main works about learning-based watermarking methods, by categorizing the methods based on the stages at which the network is used, that is: i) *single stage training*, and ii) *double stage training*.
- Followed by, we further introduce some works in a new direction, which aim to protect 3D point cloud

models and Deep Neural Network (DNN) models via watermark. We present the watermarking methods for deep neural network in terms of the function of DNN models, i) classification network and ii) image processing network.

- Afterwards, some geometric invariant watermarking methods are also summarized by dividing into three categories: feature-based, synchronization-based, and invariant moments and transform based watermarking.
- Finally, the challenges and future work are also discussed with suggested potential directions in image watermarking.

The remainder of this paper is organized as follows. Section 2 describes the properties of digital watermarking. Section 3 presents recent techniques on popular natural images based on handcraft features, and several works on two new image forms and one transmission technique are given in Section 4. In Section 5, we introduce the watermarking techniques based on various learning frameworks. Besides, we discuss watermarking techniques that are designed for protecting the point cloud model and neural network model in Section 6. After this, we summarize the geometric invariant techniques in Section 7 and give an overview discussion and summarization in Section 8. Finally, we conclude the work in Section 9.

2. Challenges for digital watermarking

Generally, the process of watermarking can be divided into two stages [33]: embedding and detection, as shown in Figure 2. The information is inserted into the carrier, so the input of embedding stage includes carrier and secret information. Subsequently, the modified carrier is uploaded upon the Internet and the user can download the media online. In the detection stage, the secret information could be recovered from the marked carrier. Mathematically, it can be described as follows

$$X^* = Emb(X, W), \quad (1)$$

$$W^* = Ext(X^*), \quad (2)$$

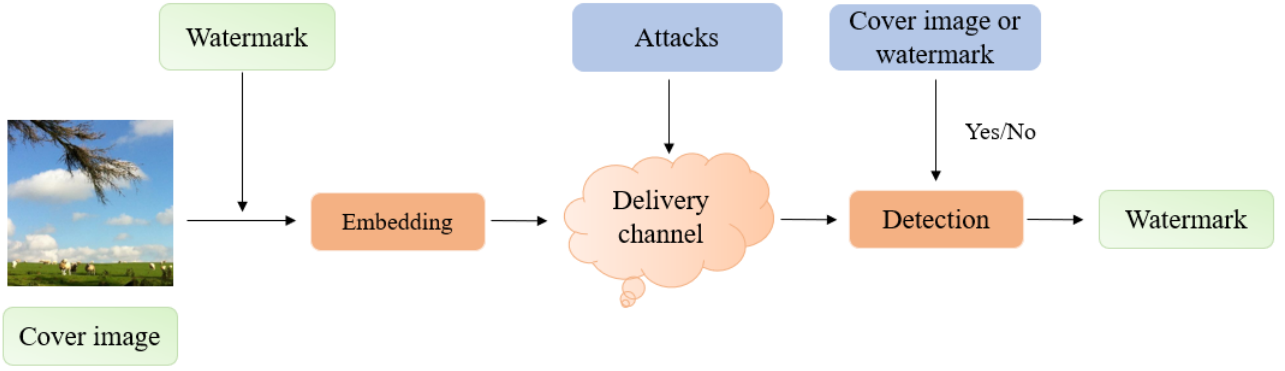


Figure 2: General flowchart of watermarking

where $Emb()$ and $Ext()$ are the symbols of embedding and extraction processing. X and X^* represent the original media and the watermarked media respectively. W and W^* are the original watermark and the recovered watermark. For a good watermarking method, some properties, such as robustness, transparency, and capacity are required and these properties usually compete with each other [34]. For instance, higher robustness is given by degrading the quality of carrier or embedding capacity. Accordingly, watermarking techniques that satisfy certain requirements could be applied to various applications, such as copyright protection [35], anti-counterfeiting of bills [36], identification of certificate authenticity [37, 38]. In this section, we summarize the properties of watermarking as follows.

2.1. Imperceptibility

The watermark embedded as additional information into the carrier media can be visible or invisible. The visible watermarking means that the watermark is regarded as one part of the media to show the ownership visibly. In the latter case, the watermark is not supposed to be visible to the naked eyes. The method satisfied with this property can greatly ensure the usefulness of host images, and avoid to be altered intentionally because the user is unaware of the existence of the watermark. Generally, the image quality assessment methods, such as mean square error (MSE), peak-signal-to-noise-ratio (PSNR), and structural similarity index model (SSIM) [39], are mostly adopted to measure the invisibility of the watermarking method. In this paper, we will focus on invisible watermarking techniques.

2.2. Robustness

Robustness means that the watermark must be invariant against various malicious or unintentional image processing operations. The robustness of a model can be measured by the extraction accuracy after various attacks. There are many ways to remove the watermark information in the digital media, which can be roughly divided into signal processing attacks and geometric attacks. Several new attacks [40–42] have also been published recently. Generally, two indices are considered to measure the robustness: bit error rate (BER) and normalized cross-correlation (NCC).

2.2.1. Signal processing attacks

Typical signal processing attacks include non-malicious and commonly used signal processing methods, such as compression, filtering, printing, and scanning [43]. We often adopt these processes on images for different requirements, such as compressing images to obtain the accelerated network transmission speed. The signal processing attack method also includes intentionally modifying the image by adding noises to attenuate the strength of the image watermark. One promising way is reversible watermarking, which can restore the original cover media through the watermark extracting process and is very useful [44–46], especially in applications dictating high fidelity of multimedia content.

2.2.2. Geometric attacks

Geometric attacks in digital watermarking systems can be divided into global geometric attacks and local geometric attacks [47]. Global geometric attacks mainly include rotation, scaling, translation, stretching distortion (shearing), and other affine transformations and projection transformations. The three transformations of rotation, scaling and translation for watermarked images are also called RST attacks, which are the most common global geometric attacks in image watermarking systems. Different from global geometric attacks on images, local geometric transformation refers to the attacks on local areas of the image. Local geometric attacks include random bending attack, cropping, row/column deletion and so on.

2.2.3. Deep attacks

With the success of deep learning in the field of computer vision and image procession, deep learning has been applied to various tasks. However, it is proved that its performance can be easily affected by adding some attentional noises so called adversarial example [48] into the carrier. Fortunately, adversarial examples have the ability to fool a trained network outputting a wrong result, but it has no effects on the non-learning-based watermarking. Nevertheless, the strong power of deep learning technologies has attracted increasing attention, and new attacks with the purpose of destroying the rules defined by watermarking are designed accordingly. Nam et al. [40] present a novel CNN-based attack for block-

based multi-bits watermarking, which finds the weak points of the targeted watermarking method and misleads the extraction by attacking the watermarked image. Namba et al. [42] present a novel attack method against watermark, i.e., query modification, and demonstrate that all of the existing watermark methods are vulnerable to either query modification or existing attack methods (model modification). To overcome this vulnerability, they further present a novel watermarking method, i.e., exponential weighting. The experimental results show that the proposed watermarking method achieves good verification performance of watermark even under a malicious attempt of unauthorized service providers, such as model modification and query modification, without sacrificing the predictive performance of the neural network model.

2.3. Other requirements

Basically, imperceptibility and robustness are required while the watermarking method is designed. In addition, some requirements may be required depending on the task.

- 1) **Capacity** The capacity denotes that the amount of watermark could be embedded in an images. There is always a balance that higher capacity will decrease the image quality and robustness of the method. Thus, it usually represents the maximum amount of watermark that can be hidden in the image [49], while maintaining a good trade-off between imperceptibility and robustness.
- 2) **Security** The security of a watermark refers to its ability to resist hostile attacks. A hostile attack is any process specifically that intends to thwart the watermark's purpose [50, 51]. If an illegal user cannot have the chance to remove the watermark without any knowledge of watermarking embedding and detector, it is considered to be secure. However, there still are some operations that can remove the watermark without the prior knowledge. For example, an unauthorized watermark is embedded into the media that has been already watermarked, so that an illegal user can claim the ownership of this media [52].
- 3) **Blind detection** It is easy to recover the watermark message from the authorized image by comparing with the original image. However, this kind of methods is hard to implement practically as the original media may not be available. To this end, it is necessary to detect and extract the message without the original image, even any auxiliary information, which has strong practicability and wide applications.

3. Watermarking for natural image

As shown in Fig. 1, the emergence of the word *watermark* can be traced back to 1993 [53]. In this section, we start providing a thorough review of the watermarking for natural images, which has been presented and developed in the past decades [54, 55]. Generally, the natural image is statistically redundant in intensity, color, high-order and other features [56], and it is verified that only a very small fraction can be visually observed by human eyes [57], which can be

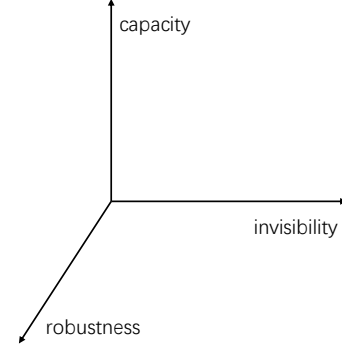


Figure 3: The trade-offs among imperceptibility, robustness, and capacity

utilized to hide much secret information for communication. Following the domain where the watermark is embedded, we introduce natural image watermarking methods in the spatial domain and the transform domain, aiming at addressing the trade-off between robustness and imperceptibility in different ways, mostly from the view of human visual system (HVS).

3.1. Spatial domain

In the early stage of watermarking technology, most works are designed in the spatial domain, which is easy to implement on computational point of view but too fragile to withstand large varieties of external attacks. Spatial domain-based schemes focus on directly modifying the pixels of images. Basically, the most famous and efficient method is Least Significant Bit (LSB), where the watermark can be embedded into an intensity image by instituting the LSB of the pixels with the watermark bit, and offers simple decoding [58]. One of the earliest spatially LSB-based watermarking schemes is introduced by L. F. Turner and R. G. Van Schyndel [59], which exhibits strong hiding ability with undetectable visual distortion. However, the watermark cannot survive if any additional operations are conducted on the image. To address this drawback, Lee et al. [60] propose to adopt the randomly embedding technique using a random mapping function with seed to reduce the possibility of watermark extracted by eavesdropper. This work shows great robustness against cropping and equalization attacks. Later, Fazli and Khodaverdi [61] investigate the trade-off between robustness and imperceptibility using the LSB method based on SSIM metric, which is more related to HVS characteristic. It exhibits that although the LSB watermarking has good imperceptibility in using of few bit-planes of the asset image, its robustness is not satisfactory. Thus, in the next few years, variants of the LSB method have been proposed to improve the trade-off performance between imperceptibility and robustness [62–66].

Dehkordi et al. [63] propose a novel digital watermarking method that utilizes SSIM to control the quality of the modified image after LSB substitution. Heidari and Naseri [64] propose a novel LSB method using Novel Enhanced Quantum Representation of digital images (NEQR) to repre-

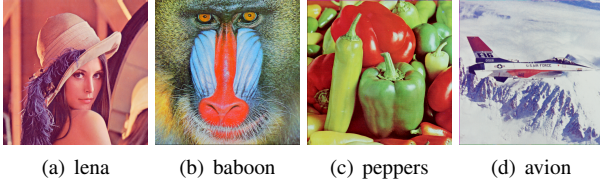


Figure 4: Classic natural images used in watermarking.

sent the original image, and then the watermark can be embedded and extracted through m -bit embedding key K_1 and m -bit extracting key K_2 . Huynh-The and Lee [65] combine MSB-LSB and 2D Otsu thresholding for color image watermarking. The gray-scale watermark image is decomposed to binary images in ordering from LSB to MSB. The optimal wavelet coefficients are quantized to represent different binary watermark 0-bits and 1-bits for better visual quality. Likewise, Kuang et al. [66] consider embedding and extracting the watermark by using a four-step phase shift and LSB. The method shows approving performance in terms of visual quality and robustness. Although LSB starts the development of watermarking, it is gradually ignored in the field of digital watermarking due to its unsatisfactory robustness against most image processing.

3.2. Transform domain

The methods designed in the spatial domain are convenient, time-saving, and robust against simple attacks like cropping and addition noise. However, the limitation of spatial-based methods is obvious, that is, it is still not robust to most of signal processing attacks and it is hard to be improved by considering the HVS characteristic. For this reason, lots of techniques have been proposed based on transform domain, including discrete cosine transformation (DCT), discrete wavelet transformation (DWT), discrete fourier transformation (DFT) and so on. In these methods, the host image is firstly transformed according to different mathematical basis functions and the watermark is inserted into the selected coefficients. The visual quality of the watermarked image can be promised because the characteristics of the HVS [67, 68] are better captured by the spectral coefficients and the information can be spread out to entire image. It is flexible to improve the performance by combining with advent visual psychology models. Furthermore, the quaternion variants of transformation domain methods, such as quaternion wavelet transformation (QWT) [69], quaternion fourier transformation (QFT) [70], quaternion discrete cosine transformation (QDCT) [71], and quaternion discrete fourier transformation (QDFT) [72] have also been explored to embed the watermark for the reason of preserving the high relationship between different color channels. Therefore, in this subsection, we make a brief overview of the watermarking methods in accordance of the transformation techniques.

DCT 2D-DCT is widely used in processing images and videos, which can be obtained by applying the 1D-DCT

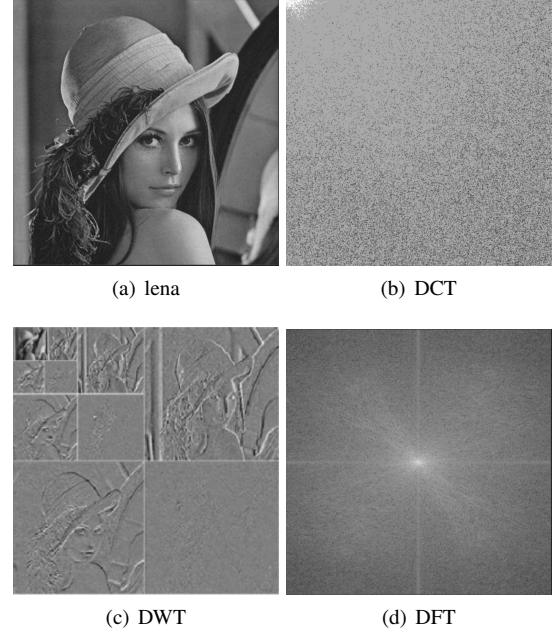


Figure 5: Three spectrum examples of lena image in DCT, DWT, and DFT domain, respectively.

along horizontal and vertical directions respectively. After DCT transformation, most of the energy of the image is included in the left-up corner coefficient, i.e., direct current coefficient, the values of the coefficients from left-up to right-bottom decrease gradually. This characteristic makes DCT be the basis of the image and video compression. It is easy to keep the compressed medias visually realistic by keeping most of the energy of the image and to improve the robustness against lossy compression. At the beginning of the study on DCT-based watermarking, most of the methods embedded the watermark into DCT coefficients directly by using a fixed embedding strength, e.g., quantization step [73, 74], for all the blocks. There are some good representative methods, such as Dither Modulation (DM) [75], Quantization Index Modulation (QIM) [76], Differential Modulation (DM) [77], Rational Dither Modulation (RIM) [78]. Although the performance of imperceptibility and robustness is much better than spatial-based methods, it is still argued that the performance can be further improved by considering visual models, which imitates the reaction of human brain while watching images [79]. Thus, with the great progress on visual physiological experiments, visual characteristic based works have been proposed [80–87].

Li et al. [80] was among the first works to combine the Spread Transform Dither Modulation (STDm) technique with Watson's perceptual model [88]. The perceptual model is utilized as a visual slack vector to get the random vector for quantization step. The results show that the proposed algorithm has good robustness against noise and JPEG compression, but it is fragile to amplitude scaling. To solve this, they further propose a robust method against this attack by using the perceptual model to get random vector and quantization

step, simultaneously [81]. Based on these works, diverse watermarking methods were proposed based on HVS characteristics to improve the trade-off between invisibility and robustness. Ernawan and Kabir [83] embed the watermark in the DCT domain with an optimal DCT-Psychovisual threshold. The gap between the psychovisual threshold and minimum quantization value of JPEG compression is utilized to embed the watermark.

On the other hand, Wan et al. exploit the relationship between Just Noticeable Difference (JND) model and watermarking method, and insert the watermark through STDM technique [89–91]. In [89], abstractly, the authors incorporate the influence of average pixel intensity into the fixed gain attacks using an invariant constant and edge intensity. The edge intensity, which is confirmed to be robust against common image processing, is replaced by three alternating coefficients. Then, the JND model is applied to logarithm STDM watermarking framework. This work, however, simplifies the feature of blocks into flat, edge, and texture, which is insufficient to measure the abundant sensory of human eyes. Thus, in [90], the pattern complexity is further considered to JND model, which digs the fact that the changes cannot be observed in a region with higher complexity compared with regular regions. The algorithm solves the problem of watermark embedding and extraction mismatch caused by anti-corruption calculation between the spatial domain and DCT domain. Furthermore, Wang et al. [91] extends this regularity by considering direction diversity and color complexity in color image watermarking. The algorithm uses the visual mask to make the effect of the surrounding content on the observation position be consistent with the process of visual perception. It extracts the direction features of the image and calculates its difference as a new neighborhood masking factor.

In addition to the JND model, Visual Saliency (VS) model, which reflects the reaction of human brain to signal stimuli and assigns attention to most important region, has been utilized to enhance the influence of JND model in watermarking process [92–94]. For instance, Niu et al. [95] first employ the visual effects of spatial VS model on JND and applies it into watermarking procedure. The most attractive region will be allocated more attention and JND value will drop. Otherwise, the JND value will increase, allowing a large modification. More recently, Li et al. [96] incorporate the DCT-based VS model to modulate the sensitivity of JND models based on the sensitivity difference of HVS to different directions to match the HVS. The experimental results show that the proposed scheme outperforms the existing technologies in terms of VSI and robustness. Nevertheless, the VS model used in this work only considers the low-level features, and it is insufficient to reflect the position of the object, which is usually compelling while someone observes the image on purpose. To imitate this mechanism for VS modeling, Wan et al. [97] proposes a new two-layer VS-guided JND model watermarking algorithm, which can better locate the profile of the object. Specifically, the top-down features from DCT coefficients of image blocks ac-

cording to the focus information of the image are integrated with bottom-up features. The experiments and oblation test of the VS model prove that the algorithm has obvious robustness and achieves good visual attention estimation.

DWT Compared with DCT transformation, DWT describes the HVS more accurate than DCT due to its multi-resolution description of the image, and is robust against the attacks like additive noise, resizing, and half toning. We may stress this for the response. The DWT decomposes the image into four subbands: the LL holds the most energy of the image and LH, HL and HH are three subbands containing the details of the image [98, 99]. The image could be decomposed into k -levels by consistently decomposing the LL channel. The higher the level, the more details information contained in the corresponding subbands. Diverse watermarking methods designed on different levels show significant difference on performance [100–102].

The widely used embedding strategy in the DWT domain is by adding watermark information to host images directly with one or two embedding strength [103, 104]. For instance, Kashyap and Sinha [103] embed and extract the watermark information into the third level LL channel using alpha blending based on two scale factors for original and watermark images. Choudhary and Parmar [104] propose a watermarking method in the 2-level DWT domain using an visibility factor k to scale the watermark image into original image. These methods can obtain higher visual quality while is not able to ensure the robustness due to the limitation of the technique itself. Thus, Li et al. [105] utilize an efficient embedding technique i.e., QIM, and embed the watermark into the significant amplitude difference in the 3rd level of DWT domain. Two smallest coefficients in the LH3 are subtracted and the result is modified as the host value. The experimental results show its good robustness against JPEG2000 compression. On the other hand, similar to the processing in DCT-based watermarking, Liu et al. [106] apply the visual saliency model and Laplacian distribution to get an adaptive image watermarking algorithm in DWT domain. An adaptive multiplicative watermark strength factor is designed in this algorithm according to the energy aggregation of the high frequency wavelet sub-band, texture masking, and visual saliency characteristics.

In addition, to further exploit better visual features in DWT domain, dual-tree complex wavelet transform (DTCWT), which can be obtained by separately filtering along rows and columns [107], has been a new way to model HVS effects [108–110]. It solves three major limitations like shift sensitivity, poor directivity example inability to distinguish between $+45^\circ$ and -45° spectral features, and absence of phase information in the conventional DWT domain [111]. Based on this, Sudha and Thanuja [109] combine DTCWT and principle component analysis for better visual quality and robustness against noise. Zebbiche et al. [110] exploit HVS characteristics, namely, the frequency band sensitivity, the brightness masking and the texture masking to quantify the amount of unnoticeable changes

in the DTCWT domain.

DFT DFT is resistant to most of geometrical processing because the magnitude keeps invariant and it can be chosen as the host to carry secret information. In the case of DFT, an image is showed as complex-valued Fourier coefficients and represented in the form of amplitude and phase, in which the most important component is the low-frequency called central component [112–115]. Gourrame et al. [113], for example, propose a robust print-cam resilient watermarking based on fourier domain. The Hough line is used to detect four corners of the image and estimate the projective matrix to correct the image before extraction. Jimson and Hemachandran [114] propose an efficient method by exchanging the DFT coefficient. They divide the image into non-overlapping blocks of 8x8 and exchange the magnitude of DFT coefficient in each block. Prajwalasimha et al. [115] propose a scheme based on the combination of DFT and Successive Division. Through DFT, better imperceptibility is noticed and the algorithm utilizes much less execution time compared with other existing approaches.

Hybrid domain Though the spatial and transform domain methods achieve better performance on invisibility and robustness, the performance of watermarking algorithms proposed in only one domain is limited in various scenarios, which should be considered carefully in specific demands. Thus, in order to obtain a satisfactory performance for watermarking application, many methods have been proposed based on hybrid domains [116–121].

Ansari and Pant [116] propose a multi-purpose image watermarking scheme to provide tamper localization, self-recovery, and ownership verification of the host image. The image is first transformed into the DWT domain and then singular values of the transformed host are modified according to different watermark bits. After the insertion, the last two LSBs of the host are modified to bear the tamper localization information, in which the LSB insertion is used to locate the tampered region as well as to provide the self-recovery ability of the scheme. Singh and Singh [117] propose a scheme to solve the most frequently occurring unauthorized reading and false positive detection security problems via combining singular value decomposition (SVD), DCT and DWT transformation. The proposed scheme is secure and free from the false positive detection problem because there is no need to chose scaling factors. Su and Chen [118] embed the binary watermark by averaging the modification of the direct current (DC) coefficient to every pixel in the spatial domain. In this way, the watermarking is executed in the spatial domain considering DC coefficient, which takes advantage of the spatial and DCT domain. Zhang et al. [119] propose a novel color image blind watermarking algorithm by integrating the unique feature of the DC component of DFT, which has strong robustness and high real time under the premise of better watermark invisibility, and solves the puzzles of slow running-speed and weak robustness. Liu et al. [120] propose a novel image

watermarking method based on DWT, Hessenberg decomposition (HD), and SVD, which enabled different sizes of watermark into host images. The fruit fly optimization algorithm, one of the natural-inspired optimization algorithms, is devoted to find the scaling factor through the proposed objective evaluation function. The experimental results show that the proposed image watermarking method has a good trade-off between robustness and invisibility even for the watermarks with multiple sizes.

Quaternion domain With the popularity of affordable digital cameras, most of the acquired images are chromatic. Compared with gray images, color images carry more information and can attain higher fidelity [69]. Basically, the blue channel can be used to hide the watermark information because it is less sensitive to human eyes, while there is a high relationship in RGB space which is not suitable for watermarking. And the Y channel of YCbCr color space is also utilized for watermarking [122]. *However, any operation on a single color channel may destroy the relation between different color channels* [123]. To preserve this property, the quaternion transformation is utilized.

The quaternion is firstly introduced by Hamilton in [124], and the main advantage of quaternion representation is that a color image can be processed holistically as a vector field. A quaternion number can be represented in the form

$$q = q_r + q_i i + q_j j + q_k k, \quad (3)$$

where q_r , q_i , q_j , and q_k are real numbers, and i , j , k are the basis elements of quaternions. These three imaginary units follow the following rules

$$\begin{aligned} i^2 = j^2 = k^2 = -1, ij = -ji = k, \\ jk = -kj = i, ki = -ik = j. \end{aligned} \quad (4)$$

In this way, a RGB color image $f(x, y)$ can be represented as a pure quaternion

$$f(x, y) = f_R(x, y)i + f_G(x, y)j + f_B(x, y)k, \quad (5)$$

where $f_R(x, y)$, $f_G(x, y)$, and $f_B(x, y)$ are red, green, and blue channels of the image, respectively. According to this representation, the color information of the image has been considered as a whole and it has been successfully used in the field of watermarking.

As mentioned above, the methods designed in one channel using frequency transformation are usually sensitive to color attacks and geometric neglecting of the watermark desynchronization. To tackle this problem, Wang et al. [70] propose a robust blind color image watermarking method using quaternion Fourier transform and least squares support vector machine (LS-SVM). The digital watermark is embedded into original color image by adaptively modulating the real quaternion Fourier transform coefficients of color image block and the LS-SVM correction with pseudo-Zernike moments is utilized for extraction. Chen et al. [69] exploit discrete fractional random transform in the form of quaternion

(QDFRNT) for color image watermarking. The embedding is conducted on the block after QDFRNT and middle coefficients are modified to represent the watermark information based on the HVS perceptual masking. The SVM is utilized to extract the watermark information. The experimental results show that the proposed method is robust to noise and cropping operation, and weak to compression processing.

More recently, Li et al. [71] embed the watermark into the coefficients after QDCT and SVD operation using an optimized encrypted binary computer-generated hologram as the watermark. The multiple embedding strength factors are adopted for watermarking using the particle swarm optimization algorithm. An improved geometric detection method is proposed to estimate the geometrical distortion parameters for extraction based on the color invariance model. The proposed method can obtain accurate geometric parameters and better performance in terms of imperceptibility and robustness against common attacks and geometric attacks. Chen et al. [123] first establish the links between the DFT of the three RGB color channels and the components of QDFT coefficients while considering a general unit pure quaternion. Then, they discuss the symmetry constraints one should follow when modifying QDFT coefficients for different unit pure quaternions i, j, k or their linear combinations. The theoretical analysis and experimental results show that these algorithms offer better performance in terms of capacity and robustness to most common attacks.

4. Embedding under sophisticated scenarios

In the parallel, with the improvement of computing ability of computer and spread of personal electrical device in our daily life, diverse images are emerging and the digital watermarking is facing new challenges under this situation. Like we said in section above, for natural image watermarking, the main solution to find a balanced tradeoff between image quality and robustness is to consider embedding watermark information in an effective domain, by combining with mature visual psychological model. However, it is still a challenge to solve specific problems in accordance of different images. In this section, we present several image watermarking methods on two new image forms, i.e., High Dynamic Range (HDR) images [125] and Screen Content Images (SCIs) [126], and a new type watermarking based on depth image based rendering (DIBR) [127], by tackling problems associating with image processing.

4.1. HDR embedding

The traditional 8-bit low dynamic (LDR) image will lose part of the scene information due to over exposure and under exposure, and result in a significant decline in viewing quality [129]. HDR imaging technology displays the rich color details and shading levels of the real world [130]. The rapid development and application of HDR in digital photography, ultra high definition television, electronic games, remote sensing detection, medical images and other aspects makes it difficult to avoid tampering and destruction in the

process of network transmission. Besides the common signal processing attacks and geometric attacks as mentioned in Section 2.2, HDR also suffers a new kind of image processing attack called tone-mapping operators (TMOs). Therefore, how to protect the authenticity, integrity and ownership of HDR images has attracted increasing attention [131, 132].

The work proposed by Guerrini is the pioneer on data hiding of HDR images [133]. It is a blind watermarking method and the capacity is single bit, which is sacrificed to favor the other requirements: noise addition, security, imperceptibility, and robustness against TMOs. Since then, several works have been conducted on embedding multi-bit embedding [134–137]. For example, the RGBe [134] and uncompressed LogLuv TIFF format [138] are considered to hide information using LSB technique, respectively. Ratanacharuchinda and Amornraksa [135] select blue component of a detail layer obtained by exploiting bilateral filtering. These methods, however, didn't conduct enough experiments on intentional or unintentional attacks. Thanks to the properties of the considered Radon DCT (RDCT) transform, Maiorana and Campisi [139] propose a robust method, which guarantees proper imperceptibility of the inserted marks, while providing the means for blindly extracting the embedded information without the need of the original image. The effects of signal processing attacks such as Gaussian additive noise, filtering and TMOs have been evaluated and it shows that this work has good robustness compared with previous works. However, there is still a huge space to be exploited against TMOs, which is necessary when HDR media need to be experienced using conventional displays in order to generate LDR data retaining as much information as possible from the original objects, while reducing the overall contrast [140].

To this end, on the purpose of making the watermark survive after TMOs, Wu [136] first make an analysis of TMOs attack in DCT domain by modifying two middle DCT coefficients per block. More recently, Bakhsh and Moghaddam [141] propose a robust HDR images watermarking method using ABC algorithm. Specially, the embedded blocks are selected using ABC algorithm and the watermark is embedded in the 1st level approximation sub-band of DWT in each block after LogLuv transformation. Considering the performance measured by BER, the proposed approach is robust against TMOs attacks and regular attacks except geometric attacks like rotating and scaling. Yu et al. [142] and Luo et al. [143] propose a robust method based on Tucker decomposition to increase the robustness to tone mapping. The first feature map of the three feature maps after tucker decomposition is employed to embed watermark, and good robustness can be achieved because the first feature map contains most of the energies in the HDR image, which is robust to various attacks. Local correlation are exploited by Auto-Regressive prediction and Schur decomposition, where the texture or edge remains after the TMOs operation. The experimental results show that the proposed method gains a significant improvement against TMOs operation, and state-of-the-art performance resistant to common image attacks.



Figure 6: Examples of HDR images [128].

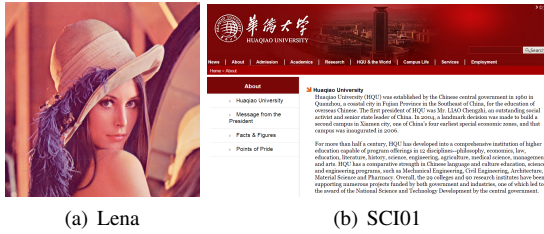


Figure 7: Samples for natural images and SCIs [144].

The results in [142] and [143] verify the superior performance of Tucker decomposition against most image processing of HDR images. A limitation, however, could be robustness against common image attacks, such as cropping and color quantization. To solve this, the effective use of human visual perception characteristics can improve the robustness while keeping image quality unchanged.

4.2. SCIs embedding

Screen Content images (SCIs) often appear in various multimedia applications and services, such as online news, online education and cloud games. The processing of SCIs, including segmentation [145], image quality evaluation [146] and compression [147], has attracted the attention of many researchers. However, most of the existing digital watermarking algorithms are designed for natural images as mentioned above. First of all, the image structure and statistical characteristics of SCIs are usually completely different from those of natural images [148]. As shown in Figure 7, generally, computer-generated SCIs with discontinuous hues have sharp edges and thin lines, with almost no color, while natural images have continuous tones, smoother edges, more colors and thicker lines [149]. Obviously, the models designed for natural images are not suitable for SCIs because the masking effects from different parts in these images have noticeable difference in human eyes [150]. Therefore, for SCIs, designing corresponding visual features to guide watermark processing of image contents will theoretically achieve better invisibility and robustness than directly applying the existing visual model to the screen content im-

age.

In order to differentiate the text region from the picture part, Wan et al. [151] propose a hybrid JND digital watermarking algorithm for screen content images based on [152]. According to the statistical difference between the image content and the text content in the SCIs, the coefficients which are not involved in the watermark processing are used to extract the features and realize the classification of the image content. Based on the classification results, the hybrid JND model is constructed, where the structural complexity is used to model the contrast masking factor of the text patch. Experimental results show that the proposed watermarking algorithm is more suitable for SCIs images compared to others natural image watermarking methods.

Certainly, a simple content classification and block-based watermarking can not provide satisfied results in this case, since the flat regions in the image has no redundant information for information hiding. The best option for SCIs could be embed watermark in the text or picture in the image, which tolerates more modification than flat parts.

4.3. DIBR embedding

DIBR is a technique to render new views from a video stream. The scene geometry is given by an additional depth stream. This technique allows for a wide variety of applications including 2D-3D video conversion [154]. Different from 2D image transmission, DIBR only transmits the depth map and center view to the receiver. The entire information can be displayed by combining these two images to generate the right and left images as shown in Figure 8. Accordingly, based on the particularity and necessity of merging different images at the end, the protection of DIBR has been exploited in recent years to resist the geometric processing [155–160].

At the very beginning, Lin and Wu [155] propose a method of embedding watermark by predicting pixel's moving distance in the DCT domain. The method provides a baseline for DIBR watermarking and various efforts are still needed. Later, Kim et al. [156] consider watermarking in DTCWT domain with quantization. The method is robust to common attacks, but it is vulnerable to geometric attack when the depth information is modified. Afterwards, in or-

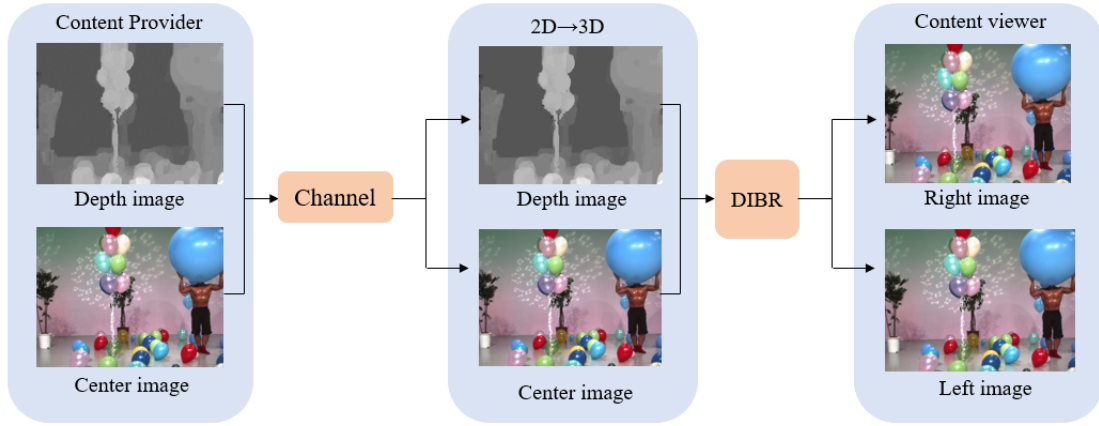


Figure 8: Framework of DIBR for generating synthesized images [153].

der to improve the performance against geometric operation, the invariant features are exploited by the scale-invariant feature transform (SIFT) to carry binary information [154, 161]. The SIFT algorithm is able to extract the matched features from two similar objects, which implies that the receiver should own the access to original image while this is limited in practice application.

For the same purpose, Chen and Zhao [157] propose a robust watermarking method based on the contourlet domain of center view to addressing the weakness of geometric robustness. Four directional coefficients after contourlet transform are chosen and the watermark information is quantized into these coefficients. The quantization error differences obtained after the embedding process are used for watermark extraction. Kim et al. [158] propose a template-based method using center images, where the template is embedded into the chosen coefficients of the template blocks, and it can be utilized to restore the synchronization error caused by geometric distortion. The extraction of watermark information is executed after synchronization of received images using the ICP technique. Lee et al. [159] propose to generate a special map from depth map, termed as the depth variation prediction map (DVPM), which is able to find locations that are safe from viewpoint changes. Nam et al. [160] propose to embed and detect the watermark in the 3-level non-subsampled contourlet transform (NSCT) domain blindly of center images, which can better eliminate the influence on the watermark after the processing of generating synthetic images and is robust to desynchronization attacks. Nevertheless, it is worth noting that the methods mentioned above were designed on one of the view images of DIBR transmission. Based the transmission rules of DIBR, however, simple watermark operation on a single carrier like depth and center view is easy to be affected because of the information sharing between right and left images. To overcome this challenge, Luo and Peng [162] combine the DIBR rendering rules to construct steady features by extracting the matched features from center and two adjacent views through SIFT. The embedding positions are robust to the affine transformation and

geometric attacks, and the robustness of the watermark is improved.

Conclusively, contourlet transform has been widely implemented for DIBR image watermarking, and most works pay attention to the resistance of DIBR processing and geometric attacks. Nevertheless, only a few works explore the robustness on combinatorial attacks, for example, rotation with translation and scaling [158]. Furthermore, it is very important to apply these DIBR image watermarking methods to 3D video watermarking in the future.

5. Watermarking with neural network

So far, diverse conventional digital watermarking methods are illustrated above in different embedding domains and for different image forms. All the methods have obtained satisfactory performance with respect to different aims. However, the common watermarking methods are devised depend on a large amount of knowledges for different types of carrier. It greatly slows down the development of digital watermarking because of the emergence of compound digital media, such as screen content media, which contains image, text, line, and any new forms from computer and many factors should be considered while applying watermark into it.

On the other hand, recently, learning-based methods attracted extensive attentions in many fields and achieves great success [163–167]. Generally, it can be categorized into machine learning and deep learning [168], where deep learning is a particular kind of machine learning that achieves great power and flexibility by learning to represent the world as a nested hierarchy of concepts. It provides the ability to learn how to make the right decision by formulating the problem as minimization of some loss functions on a training set of examples [169–171], which expresses the discrepancy between the predictions of the model being trained and the actual problem instances.

In this section, we will make a comprehensive illustration about learning-based hiding method. First of all, some

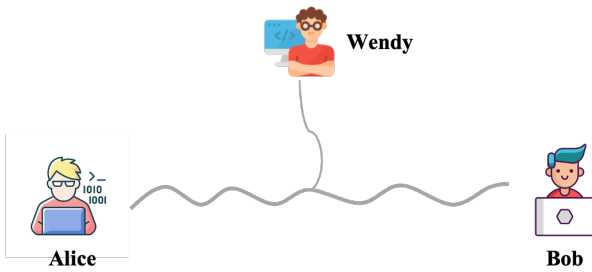


Figure 9: Simmons model of steganography. For prisoners Alice and Bob, how to make sure that the communication is covert and cannot be noticed by prison guard Wendy.

learning-based steganography schemes are illustrated. And then, we make an explanation on the efforts conducted on digital watermarking based on the learning framework.

5.1. Preview on learning-based steganography

Similar to digital watermarking, steganography is also a field of hiding information into the media, by emphasizing the concealment of confidential information instead of the protection of media itself as shown in Figure 9. To some extent, the embedding of digital watermarking and steganography is the same with different goals. Compared with digital watermarking, techniques designed for steganography have made significant progress based on deep models for various digital medias [172, 173]. Qian et al. [174] transfer the properties of the trained model from high payload detection to a lower payload detection task. Wang et al. [175] propose to detect the MP3 steganography in the entropy code domain. Using the high representation of a deep model, it is easy to reduce the visualization of hiding information in the media.

Adversarial fashion Nevertheless, the presence of secret information hidden by steganography can be detected if the attacker has knowledge of the type of data or model that is used for steganography [176]. Thus, it is inevitable to design the model assuming that there is an illegal user attempts to reveal the hiding information in an adversarial way or GAN model. To this end, several works have been executed recently in an adversarial way [177–180]. Tang et al. [177] propose a way to adversely embed the secret information into the image aiming to fool the existence of a convolutional neural network (CNN)-based steganalyzer. It has been proved that the ASDL-GAN can get a reasonable cost function for content adaptive steganography. However, this subnetwork needs to be pre-trained before the adversarial training, which is time consuming. In order to solve the weakness in [177], Yang et al. [178] propose a distortion function generating a framework for steganography to learn the embedding cost for image steganography. It designs a generator with a U-Net architecture and a no-pretraining-required double-tanh function is proposed to approximate the optimal embedding simulator. Also, based on ADV-EMB, Bernard et al. [179] propose a min max strategy to improve the security of the

iterative strategy used in [177]. It combines targeted attacks on a given detector with a min max strategy, which dynamically selects the most difficult stego content associated with the best classifier at each iteration. Furthermore, inspired by the success of building a cost map via fooling a steganalytic classifier in an iterative manner in these works, Mo et al. [180] propose a novel steganography framework based on an adversarial example in an iterative manner. The proposed framework updates the cost iteratively based on adversarial examples derived from a series of steganalytic networks until achieving satisfactory results.

Coverless communication Anyway, the steganography technology is conducted by modifying the carrier to satisfy a certain requirement, where the trace can probably be detected. Thus, in order to avoid the probability of being detected by an illegal analyzer, some works are conducted to pass the information using images without modifying these images, called coverless information hiding. Duan and Song [181] propose a coverless information hiding method based on the generative model. The secret image is put into the WGAN model database, and a meaning-normal and independent image different from the secret image is generated. Then, the receiver will receive the generated image and feed it to the model to generate another image as similar as the secret image. The transmitted images are unrelated to the secret image but able to deliver the information from the user. But it requires more trained models with the increase of transmitted information, which is time-consuming. Zhou et al. [182] propose a novel coverless information hiding method based on the faster-RCNN network. Especially, the secret is converted into binary messages at first. Then, the faster-RCNN model is utilized to detect and locate the objective of the image and the attributes of the object such as position and size are sorted to represent the corresponding binary message.

Consequently, with the development of coverless information hiding, Meng et al. [183] propose a new steganography algorithm based on object detection and relationship mapping, which integrates coverless information hiding and steganography. The coverless information hiding is realized by mapping the object type, color and secret information in object detection method. And it can not only improve the steganographic capacity but also make the coverless information hiding more secure and robust. Meanwhile, Tancik et al. [184] design a robust framework for the purpose of protecting the copyright of the images in physical photographs. The encoded image is obtained by feeding the original cover image and processed message into a U-Net style architecture. Later, an off-the-shelf semantic segmentation network BiSiNet is utilized to localize the image in the real world, and then it is used to extract the message by the decoder network. Accordingly, they define the loss with four objectives including an L2 residual regularization, the LPIPS perceptual loss LP, a critic loss LC calculated between the encoded image and the original image, and cross entropy loss for the message.

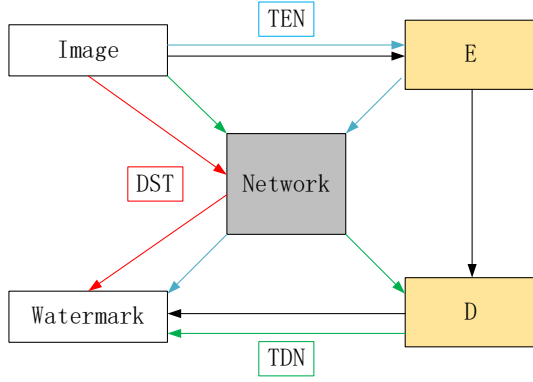


Figure 10: Brief diagram of SST and DST learning framework. E and D can be classic watermarking embedding and detection technique. Black line is traditional watermarking processing. Blue (TEN, trained embedding network) and green (TDN, trained detection network) represent that the network is trained to embed or detect watermark independently. Red line is DST, where embedding and detection are executed using trained network.

5.2. Learning-based watermarking

The basis of watermarking is that the user modifies the image with different functions, which assigns the modified value into different regions representing corresponding watermark information 0 or 1. In other words, the features of each region have the ability to represent the watermark to some extent. This is resemble to the classification task in the field of computer vision that the machine is trained to learn different features to represent definite labels automatically. Based on the discussion, extensive researches have been conducted to investigate the watermarking technique by associating with the learning framework [185, 186]. The schemes associating with the learning framework can be roughly divided into two categories: single-stage-training (SST) [187–200] and double-stage-training (DST) [201–203] as shown in Figure 10. The former refers to the methods that are designed using different neural networks for embedding or detection. The methods applying the neural networks for both embedding and detection are called DST. In this subsection, we will introduce these methods according to the types of neural networks.

5.2.1. Single-stage-training (SST)

The appearance of neural networks enables the algorithm to learn fruitful features to achieve different goals [204]. Initially, the neural network is used to learn the watermark pattern from watermarked images for detection or to learn appropriate embedding strength/location for embedding. Here we categorized these methods trained on one stage as Single-stage-training (SST) techniques.

Trained detection network (TDN) Firstly, for the detection trained watermarking methods, the watermark embedding usually leaves irreversible artifacts in the image.

Thus, some works were executed to learn watermark patterns from watermarked images by calculating the difference between the central pixel and its adjacent pixels in a given window [190–194]. In the earlier times, for instance, Yu et al. [190] embed the watermark into a pseudo random number generators (PRNGs) using amplitude modulation and a designed multilayer perceptron (MPL) network is trained to learn the difference between the original image and watermark. The signature can be restored based on the results of the learned network. Later, Zhang et al. [191] and Zear et al. [192], uses a classic back-propagation mechanism to update the weights of the network, and the wavelet domain is then selected to hide the secret information in both works.

In recent years, with the development of deep learning, especially CNN, there is much progress in the field of image processing. Compared to traditional neural networks, CNN learns the features by convolutional operation, which provides the network with a large visual field window (as shown in Figure 11) and breaks the bottleneck of the development of the traditional BP neural network. Thus, Zheng et al. [193] design a CNN architecture to extract watermark information after embedding in the DWT and SVD domain. The weights of the network are adjusted with the different watermarking images that are generated by different keys as training data. Li et al. [194] propose a novel algorithm using synergistic neural networks, where a meaningful gray watermark image is embedded into the block DCT component and extracted using a cooperative neural network.

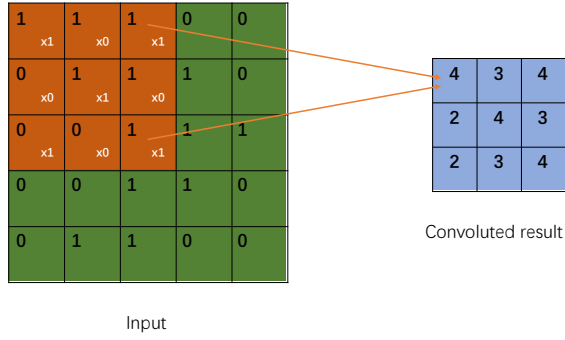
On the other hand, inspired by the success of deep learning for the image classification task, Vukotić et al. [196] discuss the ability of classification deep networks for blind watermarking based on zero-bit watermarking. The watermarking is implemented in the feature domain and the final marked image is iteratively modified by considering the concept of adversarial samples. The SSIM heat map is utilized to preserve the image quality while embedding. In addition, the image is preprocessed by using local transform to improve the robustness to geometric attacks. Afterwards, Vukotić et al. [197] continue executing the experiments with different model structures e.g., RMAC and WELDON, and achieve good performance. These methods, however, still focus on embedding the watermark conventionally, while ignoring the learning ability of neural networks from the data automatically. Furthermore, Huynh-The et al. [195] propose to extract watermark as a semantic segmentation work with convolutional encoder-decoder network. Specifically, the watermark is inserted into the DWT domain and the network is utilized to recovery watermark from the embedded maps with the size of 64×64 . It represents the difference between the embedded maps and the coefficients in the third layer of the marked image after wavelet transform.

Trained embedding network (TEN) In stead of detecting the secret message by the model, several works were also proposed to embed watermark with the help of the trained neural network [198–200]. By this way, the model can provide a visual-like embedding strength or modification for

Table 1

Summary of SST learning-based watermarking techniques. E: Embedding; D: Detection; CED: Convolutional Encoder-Decoder; CE: Cross Entropy.

Method	Trained in	Architecture	dataset	Input	watermark	objective function	Invisibility		Robust
							PSNR	SSIM	
Yu et al. [190]	D	MPL	-	480x512	32x32	-	46.10	-	Robust to sharpening, rotation, JPEG, scaling, multiple attack
Zhang et al. [191]	D	BPNN	-	512x512	64x64	-	38.05	-	Robust to JPEG, resampling, blurring, sharpening
Zear et al. [192]	D	BPNN	-	512x512	256x256	-	43.88	-	Robust to gain, JPEG, noise, resize, filter
Zheng et al. [193]	D	CNN	-	512x512	512x512	-	38.35	-	Robust to rotation, JPEG, filter, blur, contrast adjustment
Li et al. [194]	D	SRCNN	-	-	-	-	49.10	-	-
Huynh-The et al. [195]	D	CED	BOSSbase [205]	512x512	64x64	MSE	47.85	0.9948	Robust to rotation, JPEG, filter, scaling, crop, noise
Vukotić et al. [196], [197]	D	VGG19	CLIC [206]	-	-	CE&MSE	42	-	Robust to noise, JPEG, rotation, crop, gamma correction
Kandi et al. [198]	E	AE	-	128x128	64x64	MSE	58.9	0.999	Robust to noise, JPEG, filter, rotation, crop
Sy et al. [199]	E	CNN	USC-SIPI [207]	512x512	32x32	MSE	42	0.986	Robust to noise, JPEG, rotation, crop
Bagheri et al. [200]	E	Mask R-CNN	COCO [208]	512x512	4x4	-	49.1052	0.9985	Robust to noise, JPEG, histogram equalization

**Figure 11:** Convolutional operation of CNN.

the embedding. For example, Kandi et al. [198] represent the watermark information using different generated images from the same neural network assuming that the two images are visually indistinct but have different pixel values from the network. The watermarked image is reconstructed using the corresponding outputs from two networks by comparing the Euclidean distance between the watermark image blocks with the help of the original carrier image. Sy et al. [199] pre-train the model to get a novel pattern which can be regarded as the marked type of the input. Then, the original coefficients are substituted by the output of the model according to the watermark information. Similarly, the message is extracted with the help of the original image using the output of the model from the received image. Bagheri et al. [200] train the model based on the defined table that consists of two classes: person and car from the COCO dataset, and calculate the embedding strength by scaling the maximum value of the product of classes, i.e., car and person. The embedding is conducted on the DCT coefficients of 2-level DWT transformation of the blocks guided by the embedding strength. The watermark can be extracted by comparing the value of the modified coefficients.

5.2.2. Double-stage-training (DST)

For SST watermarking methods, we can see that they only focus on improving the performance for embedding or detection using neural networks. Different from the SST learning watermarking, some efforts have been conducted to mark the digital media freely guided by the expected neural networks [186, 209]. As mentioned in Section 2, an integrity watermarking methods is composed of embedding and detection based on given mathematical function. However, on the other hand, a good learning-based method is determined by the architecture of the model and predefined loss function according to different tasks. Thus, in this part, we briefly classify the learning-based watermarking methods according to their contributions, by the choice of embedding model, detection model, loss function, and attack training as shown in Figure 12.

The process of watermarking is regarded as a visually binary classification task guided by one single model with a given objective function. In the embedding process, the input image block is iteratively modified according to the result of the model, while fixing the weight of the model, to represent the watermark information. Firstly, Mun et al. [201] propose a blind color watermarking based on convolutional neural networks to substitute the QDFT and QIM as shown in Figure 12. In details, the proper modification of the block is calculated via using stochastic gradient descent, and then the changed block is fed into the model again in order to find the best domain for watermark representation several times. After that, the marked image will be attacked and input into the network for the classification task again to achieve the watermark extraction. The robustness and accuracy of the model will be improved during this attack and model updating procedure. Finally, the performance of the model can be enhanced by repeating the embedding, attacks, and extraction stage with the updated model. The performance against common image attacks, such as filtering, noise, JPEG compression, and affine transformation, is showed in the experiments. However, the back propagation for image modifica-

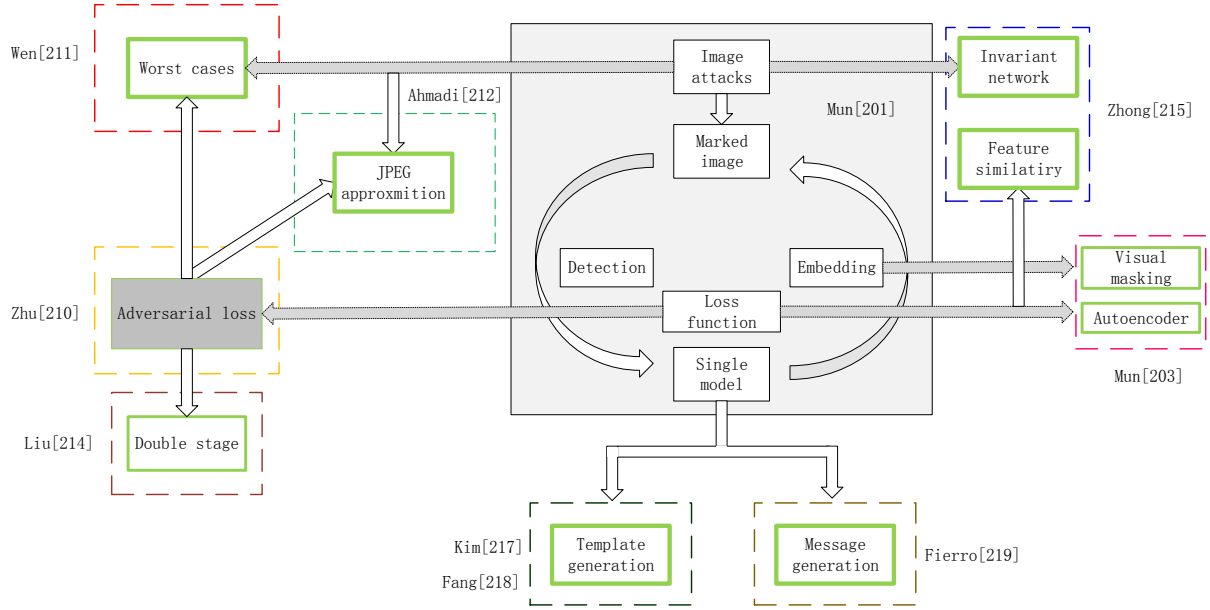


Figure 12: The main contributions of DST watermarking methods. Based on the deep insight of the works, we categorize the watermarking processing into four parts: embedding, noise, detection, and the used model, based on the first work in [201]. Specifically, we highlight the main idea of the proposed DST watermakings while ignoring the difference among the network models, as which is designed for specific aims.

tion will decrease the performance of the model and on the basis of the above work, Mun et al. [203] further improve the method based on autoencoder-decoder network. Here, the back propagation is replaced by an autoencoder network, in which the image block is altered forwardly. Meanwhile, in order to improve the quality of the watermarked image, they design a novel visual masking weight. Simultaneously, they test the performance of the model via different attacks and it shows a good visual quality than that of [201].

The methods presented by Mun have good performance in terms of invisibility and robustness. However, there is a hidden danger that the model may recover the watermark from the original image and thus reduce its reliability. To address this potential issue, several works have been done by considering the adversarial process in the objective function [210, 211, 169, 212], where the conception of adversarial process is generally used in GAN model [213]. The main idea of the GAN is that the generator is used to synthesize the fake data while the discriminator aims at telling apart the fake and real one. The two networks are updated in an adversarial way until the discriminator cannot differentiate the fake image from the dataset. Thus, Zhu et al. [210] introduce the adversarial training into the watermarking process with four parts: encoder, noising, decoder, and adversary. The encoder receives the original image and watermark message and then outputs a marked image that contains the watermark without visual distortion. The decoder handles the marked image and extracts meaningful features to construct the watermark message. Besides, an adversary network is utilized to judge whether the image contains the watermark message and improves the quality of encoded image. As the image

will be contaminated while delivering online, the authors insert several noise layers between the encoder and decoder which can enforce the model to learn robust features surviving during the delivery of image. Similarly, Liu et al. [214] consider the noise layer in a two-stage deep watermarking scheme which separately trains the model to find the best embedding and detection models. In the training stage, noising processing is not considered, and the encoder module is mainly updated to obtain a high-quality marked image. After that, the parameters of the encoder are fixed, and the marked image is fed into the decoder network after noising attack to update its parameters, so as to enhance the robustness of the model. Extensive experiments demonstrate that the proposed framework not only exhibits better stability, greater performance, and faster convergence speed than the current state-of-the-art OET methods but is also able to resist high-intensity noises that have not been tested in previous works.

For these works, the training of the model and image attacks are conducted with neural network and basic image processing library, respectively. Inspired by this, Ahmadi et al. [212] propose a transform-based deep watermarking with neural network, which combines the watermarking and image attacks together using neural network. The input image is first transformed into the frequency domain using convolution layers to enhance the robustness of the algorithm. During the training, a processing mechanism JPEG compression is constructed by the convolutional layer, through which image attacks and the update of the model will be included in one training loop.

Nevertheless, the performance seems good but one can notice that the noise attacks are fixed during training. It

Table 2
Summary of DST learning-based watermarking techniques.

Method	Architecture	dataset	Input	watermark	objective function	Invisibility		Robust
						PSNR	SSIM	
Mun et al. [201]	E and D: 14-layers CNN	BOSSbase	512x512	64x64	L2&CE	39.2	-	Robust to Gaussian Noise cropping
Mun et al. [203]	E: Ae-based D:12-layers CNN	BOSSbase	512x512	64x64	L2&CE	45	-	Robust to Row and column removal affine transform
Zhu et al. [210]	E: AE-based D:6-layers CNN	COCO	512x512 128x128	52&30	MSE&Adversarial loss	-	-	Robust to dropout,cropout JPEG, Gaussian Noise
Wen and Aydore [211]	Network in [210]	COCO	-	30	MSE&Adversarial loss	-	-	Robust to crop, dropout JPEG, Gaussian blur
Liu et al. [169]	E:7-layers CNN D:9-layers CNN	CIFAR10&COCO	128x128	30	MSE&Adversarial loss	33.5	-	Robust to dropout,cropout JPEG, Gaussian filter
Ahmadi et al. [212]	E:8-layers CNN D:6-layers CNN	CIFAR10 PascalVOC COCO	32x32	4x4	CE&SSIM	40.24	0.987	Robust to JPEG, Gaussian noise mean smoothing filter
Zhong et al. [215]	E and D: 45-layers CNN	COCO&CIFAR10	128x128	32x32	L1	39.72	-	Robust to JPEG, Gaussian noise cropping, phones scan
Luo et al. [216]	E:AE-based D: -	MSCOCO	128x128	30	L2&adversarial loss	33.7	-	Robust to saturation, hue, resize noise, GIF encoding
Kim et al. [217]	template-based CNN	BOSSbase Middlebury Microsoft Research 3D Video	512x512 1800x1500	32	MSE	43.6668	0.9859	Focus on geometric attacks
Fang et al. [218]	template-based CNN	ImageNet	512x512	30	CE&MSE	39.7291	-	Focus on geometric attacks
Fierro-Radilla et al. [219]	E and D: 13-layers CNN	-	300x300	-	-	33.15	-	Robust to JPEG, filter noise, rotation

means that the model is robust to the known image attacks, but may suffer from poor performance to unknown attacks. In order to improve the robustness of the model, Wen and Aydore [211] add a detection network to automatically find the attack parameters under the condition of the worst case, and uses the marked image to recover and extract the watermark information. Experimental results show that the proposed method outperforms existing methods [210] when it is trained with the combination of all attacks, and it also avoids the over-fitting problem while training with specific attack. Later, Zhong et al. [215] attempt to map the watermark information into the feature space and enhance the robustness through an invariant network before extraction. In order to ensure the robustness of the model to recover the watermark information when the image is attacked. The image is converted from 3 channels to N channels by using the fully connected network layer and sparse activation, which eliminates the redundant information in the image to a certain extent, ensures the restoration of the watermark information, and improves the computational efficiency. Finally, the n-channel feature map is used to recover and extract the watermark. What's more, recently, Luo et al. [216] propose a distortion agnostic framework to enhance the robustness without any prior knowledge on the type of image distortions during training. To achieve this, one adversarial example is generated during the training with CNN perturbations to replace the distorted images and is fed into the decoder to recover the original message. The adversarial training part is able to find the weak link of the model and force the model to overcome the weakness during the training. The proposed method has been proved to obtain comparable results compared with the state-of-the-art and has good generalization to unknown distortions.

Geometric invariant Compared with the interests in common image processing like noise, JPEG compression and cropping attacks, the invariant performance to geometric attack attracts less attention. Recently, some new works have been presented on how to get robustness against geometric processing [217, 218]. Kim et al. [217] propose a novel learning-based scheme for recovering watermark synchronization based on CNN. They use a template generation network to generate a specific form of noise that can be detected in the template extraction network. In the stage of extraction, the template is firstly extracted by the extraction network and then used to estimate the RST parameters of the geometrically distorted image. Finally, the watermark can be decoded from the estimated watermark matrix. Likewise, Fang et al. [218] use two templates of embedding and locating to embed the watermark information and recover the distorted images. They design a symmetric flip-based locating template by repeatedly embedding the watermark and reducing the size of the template to the 1/4 size of the watermark unit. The two templates are inserted into the channels of R and B respectively. As for extraction, a coarse pattern is firstly extracted from the distorted images with the help of locating template and fed into the two-stage extracting network which consists of two subnetworks, i.e., enhancing subnet and classification subnet, for the purposes of refining the coarse pattern and recovering the watermark matrix.

Zero watermarking Recently, Fierro-Radilla et al. [219] explore zero-watermarking, in which the model is first trained and the output of the fully connection layer is utilized to generate the shared message for the master, combining with the permuted watermark information. Basically, the watermark can be recovered by feeding the images into the

pre-trained model and derived with the similar operation of shared messages. The experiments are conducted on the attacks of compression, noise adding, rotation and filtering, and achieve an average performance of NCC at 0.95.

6. Model watermarking

With the development of multimedia technologies, the carrier that should be protected is no longer limited to image, video, text, and audio. As a universal digital product, 3D models are usually stored on hard disks or other external devices in a digital form, which makes the copyright protection of 3D models very urgent. Although the application of deep learning on digital watermarking has achieved great development in recent years, there is a significant weakness in the models learned by deep learning. That is, the models can be easily fooled if the input is contaminated, which is usually called adversarial example. In some situations, people who have no right to use the models or digital media could train their own model once they have sufficient available data and the models could be tampered through backdoor or model-reuse attacks [220]. Thus, it is also urgent to protect the copyright of the model rather than the conventional digital media [221]. In this section, we make a brief review on watermarking for point cloud models and deep neural network models.

6.1. Point cloud model embedding

With the continuous improvement of 3D scanning technologies and computer model processing data performance, 3D data model has also become one of the four major multimedia data types (image, video, audio, 3D model), which is mainly used in medical imaging, cultural relic protection and display media, etc [222]. As an important form of 3D models, the protection of point cloud models has attracted the attention [223–227] and it requires the robustness against the attacks such as similarity transformation, reordering, simplification, adding noise vector to each point and cropping.

Qi and Xie [223] propose a robust point cloud model watermarking based on self-similarity in the spatial domain. In their work, the 3D point cloud model is divided into several patches using the octree structure and PCA preprocessing. Then the similarity between different patches is calculated and the similar patches are clustered to the patch chains as a codebook. In order to increase the robustness, the watermark is embedded into the local vector along the local vector direction of a certain points of each patch repeatedly. In the stage of extraction, the known model center, the principal object axis of the original model, the edge length of the bounding cube and the codebook are considered. Dong and Zeng [224] embed the watermark in the image form of the 3D point cloud model. DWT, DCT and SVD are used on the two-dimensional images and watermark/to-be-embedded watermark. Liu et al. [225] select the vertices with mean curvature less than zero as feature vertices for watermark embedding which can guarantee imperceptibility. Otherwise, the rest vertices are utilized to resist the geometric attacks, in which the 3D model is divided several ball

rings according to the number of watermark bits. The radial distances of the feature vertices are used to represent the watermark within different ball rings. In a word, the 3D point cloud watermarking has obtained a good visual quality, but it still needs to enhance the robustness in the future.

6.2. Neural model embedding

Embedding disseminative information in the deep model has drawn much attention recently with the fast development of the deep learning technology [228–236], including both black-box and white-box approaches. The black-box approaches have wider applications than the white-box ones, as their model weights are transparent to the verifier during watermark verification. Different from watermarking on images, a good neural network watermarking requires the robustness against model fine-tuning, network pruning, watermark overwriting, and surrogate model attack. In the following, we will briefly introduce some neural network watermarking works based on the task of neural networks carried on.

Classification network Initially, attempts about model watermarking were conducted on neural networks for the task of classification and universal application. Uchida et al. [228] and Nagai et al. [229] firstly propose the general white-box CNNs watermarking by embedding a watermark in model parameters using a parameter regularizer, which does not impair the performance of networks. In the experiments, they demonstrate the robustness to general model processing like fine-tuning, transfer learning, and model compression. Nevertheless, it should be noticed that the watermarking capacity is limited by the model scale and the watermark can be easily overwritten by fine-tuning with another watermark information. Besides, Rouhani et al. [230] propose to protect the ownership of deep learning models in a different way. Especially, they generate a set of N-bit binary random strings firstly and embed it in the probability density function of the data in the target network. In addition, specific input keys are used to trigger the corresponding WM strings after watermark embedding. The model is fine-tuned with particular constraints according to the watermark information within intermediate activation maps of the target DL model. In the stage of extraction, two different settings are deployed for bot situations of white-box and white-box attacks. Thus, it has been proved to be strong against overwriting attacks.

Compared with the white-box methods, the black-box ones are blind to model weights during watermark verification. The idea of black-box methods is to learn a specific relationship between specific inputs (called trigger keys) and expected outputs (called verification keys) and verify the watermark by checking whether the relationship exists. Le Merrer et al. [232] firstly consider the adversarial examples as trigger keys and fine-tune the model to classify the adversarial examples closed to their label as the verification keys. Whereas, the adversarial examples are unstable where the embedding maybe fragile to fine-tune and model compression. Following this, in order to overcome these issues, Adi et al. [237] utilize the images with triggers and the corre-

sponding key labels to retrain the cover model. They present possible attacks that are both black-box and grey-box in the model, and show how robust the proposed watermarking approach is to them. At the same time, the authors outline a theoretical connection to the previous work on backdooring such models.

Later, Zhang et al. [233] try to explore the intrinsic generalization and memorization capabilities of deep neural networks to automatically learn the patterns of embedded watermarks. They investigate three DNN-applicable watermark generation algorithms by proposing a watermark implanting approach to infuse watermark into deep learning models, and designing a remote verification mechanism to determine the model ownership. By extending the intrinsic generalization and memorization capabilities of deep neural networks, it enables the models to learn specially crafted watermarks at training and activates with pre-specified predictions when observing the watermark patterns at inference. The proposed framework accurately (100%) and quickly verifies the ownership of all the remotely deployed deep learning models without affecting the model accuracy for normal input data. In addition, the embedded watermarks in DNN models are robust and resilient to different counter-watermark mechanisms, such as fine-tuning, parameter pruning, and model inversion attacks.

Image processing network Initially, attempts about model watermarking were conducted on neural networks for the task of classification and universal application. Uchida et al. [228] and Nagai et al. [229] firstly propose the general white-box CNNs watermarking by embedding a watermark in model parameters using a parameter regularizer, which does not impair the performance of networks. In the experiments, they demonstrate the robustness to general model processing like fine-tuning, transfer learning, and model compression. Nevertheless, it should be noticed that the watermarking capacity is limited by the model scale and the watermark can be easily overwritten by fine-tuning with another watermark information. Besides, Rouhani et al. [230] propose to protect the ownership of deep learning models in a different way. Especially, they generate a set of N-bit binary random strings firstly and embed it in the probability density function of the data in the target network. In addition, specific input keys are used to trigger the corresponding WM strings after watermark embedding. The model is fine-tuned with particular constraints according to the watermark information within intermediate activation maps of the target DL model. In the stage of extraction, two different settings are deployed for bot situations of white-box and white-box attacks. Thus, it has been proved to be strong against overwriting attacks.

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7. Geometric invariant watermarking

So far, we have made a detailed description for robust watermarking methods, based on the choice of the cover image/model and embedding algorithm. It can be noticed that, through the review, issues about geometric attacks, such as rotation, scaling, cropping [238, 239] is the most challenged problem for both image and model watermarking. And it is open because minor geometric manipulation could miss the synchronization information and disable watermark detection. Thus, in this section, back to classic natural images, we present the works that make a great effort to improve the robustness against geometric attacks. More specifically, existing solutions to resisting geometric attacks are roughly categorized into three principal classes: invariant transform domain-based algorithms, feature-based methods, and synchronization-based methods.

Invariant moments and transform based watermarking The invariant moments and transforms, such as Zernike Moments (ZMs), Pseudo-Zernike Moments (PZMs), Tchebichef Moments (TMs), Complex Moments (CMs), and Polar Harmonic Transforms(PHTs), are the most popular techniques to achieve resilience against general geometric attacks. In the literature, most of the methods ex-

exploit PHTs [240–244], i.e., the Polar Complex Exponential Transform (PCET), the Polar Cosine Transform (PCT) and the Polar Sine Transform (PST), which is a kind of orthogonally invariant moment defined on the unit circle [245]. Qi et al. [241] utilize the Polar Linear Canonical Transform (PLCT), with parameters in the real special linear group to embed the watermark information. They employed the approximated ZOA method in performance simulations where the performance of the PLCT was compared with those of PHTs, ZMs and PZMs. The performance of PLCT is dependent on the selection of the parameters. That is, the inappropriate selections of parameters may result in poor performance.

Wang et al. [242] use the optimal NSST, which can provide nearly optimal approximation for 2D images, to embed digital watermark. And PHTs are exploited for estimating the geometric distortion parameters in order to permit watermark extraction. Also, they exploit the ability of the FSVM machine learning technique to estimate the geometric distortion parameters for watermark extraction. The experimental results show that the proposed approach conquers the weakness of the geometric correction-based watermarking scheme. Hosny and Darwish [243] use highly accurate and numerically stable PHTs' moments in order to provide good visual imperceptibility and high robustness against the geometric distortions. Xu et al. [244] further exploit the potential of the PHT domain, in which invariant properties of PHTs' moments are used to compute orthogonal moment features. And steady moments that are not sensitive to geometric attacks are picked out. Next, the watermark is embedded using quantization techniques and extracted with the help of original images. Besides rotation and scaling processing, it shows that the proposed method is robust to filtering, noise, and compression processing. Although the remarkable performance against global desynchronization attack can be achieved, a weakness of resisting local desynchronization attack still needs to be improved.

Synchronization-based watermarking

Synchronization-based algorithms rectify geometric distortions before watermark extraction [246]. The geometric parameters are estimated firstly using an exhaustive search or template insertion. Then, the distorted image could be recovered based on the estimated parameters and the watermark can be extracted correctly. Wang et al. [247] present a method that the watermark is embedded in the non-subsampled shearlet transform (NSST) domain and uses quaternion polar harmonic transform (PHT) and least squares support vector regression (LS-SVR) for synchronous correction. The method outperforms the previous works but has a high computational cost. Image registration overcomes this problem by using a registration algorithm to align the watermark before extraction, which is then used to recover the distortion parameters that appear in the watermark image due to geometric attacks [248]. On the other hand, a template is embedded into the original image as a supplementary to estimate the distortion parameters

before extracting the watermark [217, 218, 249]. However, it is difficult to recover the distorted images without original images or the template. In the real world, the original images are usually restricted. And the template can be easily removed by the attacker causing a failure in recovering the watermark information.

Feature-based watermarking The invariant-based methods can solve the problem of watermark desynchronization, which extracts invariant features for embedding and detecting the watermark [154, 250–253] without synchronization errors. Wang et al. [254] utilize the Harris–Laplace detector to extract steady feature points from the host image. The local feature regions (LFR) are ascertained adaptively and normalized by an image normalization technique. Finally, they embed the digital watermark into the non-overlapped normalized LFR according to the pre-distortion compensation theory. The performance of the proposed method is demonstrated against geometric attacks and common signal processing attacks.

Zhang et al. [255] combine SIFT, SVD and all phase bi-orthogonal transform (APBT) to extract a large volume of feature points for embedding. The proposed method shows great performance in terms of perceptibility and robustness on cropping. Fang et al. [256] propose an intensity-based scale-invariant feature transform (I-SIFT) algorithm which can accurately locate the embedding regions. The watermark is repeatedly embedded into different regions with a light-weight algorithm. The extraction is conducted according to a cross-validation-based extraction algorithm, which has the ability to copy with repeated embedding. The experimental results show that the proposed method is robust for the screen-shooting process. Feature-based algorithm, however, always has lesser watermark capacity restricted by the feature point and higher computation complexity for process variant features.

8. Discussion and future trends

Digital watermarking has been applied to various fields for security protection and achieved great success by combining with most of mature techniques. Due to the requirements for invisibility, robustness, and capacity, various watermarking methods have been proposed. This paper gives a survey of conventional schemes and the neural network-based schemes. Although significant progress have been made for watermarking, there are still many challenges to be tackled [257].

As the invisibility and robustness are two important metrics to evaluate the watermarking method, the studies in this field have been conducted to exploit various embedding domains and confidential HVS-simulated features. As mentioned above, different domains have their own advantages on specific aspects. The methods designed in spatial domain is time-saving but weak to most of the attacks, while transformation-based watermarking methods could overcome this problems with satisfied robustness. Nowadays, most of the works achieve good trade-off performance

in terms of invisibility and robustness, but the features used in embedding and detection are becoming more and more complicated, which means that the performances are obtained by sacrificing the time consumption. This is still a challenge for practical application and needs further significant works. Some works were tried to embed the watermark information in the spatial domain with the prior knowledge of transformation [118, 119]. It takes advantage of both the spatial domain and transformation domain.

As the HVS has higher sensitivity to luminance distortion than chrominance distortion, most of the watermarking methods are designed in luminance channels because there are numerical visual models conducted on luminance distortion which could be used for the watermarking. However, the correlation of image channels should be considered jointly. Although significant effort has been made in the quaternion domain, there is still space for further improvement on its performance in relation to compression processing. Besides, it is usual to evaluate the quality of the watermarked image using PSNR or SSIM. However, it is not able to reflect the true perceptual quality of the HVS, especially for some new image forms. Some new image quality assessment methods should be considered to assess the image quality as HVS, like VSI [258], which takes the visual attention into consideration [259, 260].

Over the past years, the development of digital technologies leads to a high resolution of digital media, such as HDR images, which requires increasing storage. To reduce the payload of digital media, the redundancy information is discarded as much as possible during the transmission [261]. For instance, restricted by the low resolution computer monitors, the details of HDR images may not be displayed completely, which facilitates the development of TMOs. This is the biggest challenge for HDR watermarking and attracts increasing attentions in recent years [141]. The existing works mainly consider the characteristics of luminance components while the correlations of RGB channels are omitted. Thus, the correlations of three color channels should be utilized sufficiently and the correlations will thus be preserved after TMOs processing. Although few works have been proposed to embed watermark using Tucker decomposition that can utilize the correlations of different channels [142, 143], the performance against TMOs still needs to be improved. Furthermore, for DIBR watermarking, most of the works focus on embedding watermark in the depth map or center image, which is proved to be fragile to warping operation during the transmission of DIBR. Some works have been taken to exploit invariant features from all the view images using SIFT algorithm [154, 161]. But it still cannot ensure the robustness against depth-map attacks and how to convert DIBR image watermarking methods to 3D video watermarking is worth exploring [162].

In the past years, different types of digital medias, such as HDR, DIBR and SCIs have emerged. The common features derived from the spatial domain and transformation domain, based on the characteristics of HVS, are still on the main research direction towards solving the challenges.

For example, SCIs contains natural images, computer generated graph, and texts. All the features of these contents should be considered separately, which requires to classify the types of the image firstly. This will be difficult because we need to make both the classification methods and visual features for embedding robust to common image processing. Also, there are some new compression strategies because this type of images has more complicated features than other images [262], which casts a great challenge for watermarking. One of the promising options for this task is deep learning, which is capable of learning the significant features and achieving the objectives driven by the defined objective functions.

Meanwhile, the availability of large amounts of data and computational resources encourages the development of ever more complex and valuable models [263]. These models need to consider the intellectual property of the legitimate parties who have trained them, which makes their protection against stealing, illegitimate redistribution, and unauthorized application an urgent need. The emergence of numerous watermarking schemes and attacks is pushed forward by academia and industry. Significant efforts have been made for modeling watermarking by exploiting the techniques of reversible watermarking, steganography for tasks of denoising, super-resolution. But there is room for future work techniques. First of all, existing works mainly focus on classification networks. Although some works have been proposed and the architectures are usually constructed based on several well-known networks, e.g., GAN, ResNet, the protection challenges of the models in image processing tasks remains. Also, the scalability of the existing schemes is also challenging due to the small size of the dataset used in the current works.

Deep learning and watermarking Learning-based watermarking schemes have shown a strong ability of fitting to different demands. Considerable effort has been made to detect the watermark information using machine learning, or neural networks driven by loss functions. However, the practicability should be considered for the practical application in the real world. First of all, for deep learning models, a large training dataset is still necessary. Training with a small dataset would be definitely another future issue for industry application. The fact is that, usually, the model is able to extract deep semantic features from the training dataset, which may not be applicable to another dataset. Thus, the transferability of deep models will be a promising direction for watermarking. Also, for learning-based watermarking, adversarial example is a huge threat, which has been proved and utilized to weaken the performance of deep models. One direction for learning-based watermarking methods is to improve the ability of the models to embed and detect the unstable watermark, in which the watermark could be removed by re-training the model on clean datasets. This could be addressed with the use of the game theory, where the protector and attacker fight with each other to improve their own performance.

Besides, the neural networks are utilized to embed and extract the watermark information aiming at keeping the invisibility and enhancing the robustness to common image processing methods. However, both conventional and learning-based watermarking methods only focused on the differentiable distortions. While the computer vision tasks, such as image enhancement, image style transferring, deepfake, have dramatically changed the way to generate or modify the image, which offers an opportunity for the user to transfer the image to completely different image content. It throws the threat to destroy the existence of watermarking when the protected images are used to train a model to generate fake images, especially when it is deliberate to remove the watermark [264–266]. For instance, Li et al. [264] proposed a visible watermark removal framework based on the conditional generative adversarial network. Here, we give two considerations on watermarked images in computer vision applications.

In general, the first choice is to prevent the damage of the watermark in the image. One option is to devise watermarking methods under the assumption of deep convolutional operation. It seems easy to train a good watermarking method based on learning frameworks. In this case, Wang et al. [267] present the DeepTAG method in order to verify the images from being used to generate a deepfake image. On the other hand, it is a good way to prevent the reuse of watermarked images for task training. Several works have been proposed, where adversarial examples were added to the authorized image datasets [268, 269]. In the case that the protected image datasets are used to train a deepfake generation network, the outputs of the networks show unacceptable visual quality. This idea could be borrowed to disturb the training of other neural networks with regular watermark information.

Beyond watermarking, inspired by impressive works on deepfake detection, it has demonstrated that neural networks working on learning features within a given color space show unexpected artifacts in other color space [270, 271]. Although each convolution layer outputs the sum of different channels, the interrelationship of the RGB channels is still not used sufficiently, and result in the loss of feature information. This could be utilized to discover the existence of watermark or even remove watermark. Thus, for deep learning models, it is still an important issue to consider color interrelationship jointly. Quaternion representation seems to perform reasonably well and few attempts have been made in the classification task [272, 273]. In [272], the color images are circulated and processed in the form of a quaternion matrix in the QCNN, which not only preserves the vertical relationship between the features, but also maintains the horizontal relationship of the color channels.

9. Conclusion

In this paper, we make a comprehensive review on digital watermarking. First of all, we give a general introduction to digital watermarking in terms of properties and classification.

Secondly, traditional techniques are categorized according to the embedding domains for natural images. And then, we present some new works for two new images and DIBR system. Followed by, we discuss the state-of-the-art techniques that are designed using various learning frameworks. And new watermarking schemes for model protection, i.e., point cloud model and DNN model, are also discussed in this paper. We also classify the geometric attacks into three categories with a comprehensive discussion. Finally, we conclude the discussion and highlight interesting future trends in digital watermarking.

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